

# Aiden Luo

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## EDUCATION

### University of California Santa Cruz

Expected Grad: June 2028

*B.S. Robotics Engineering*

*GPA : 3.78/4*

- Coursework: Computer Systems/Assembly Language, Programming Abstractions in Python, Robot Automation, SOE Calculus III, Ordinary Differential Equations, Logic Design, Applied Discrete Math, Intro Probability Theory

## EXPERIENCE

### IPUSMOBOT (Jiazhi Technology)

July 2026 - Sept 2026

*Research Development Software Engineer Intern*

*Hangzhou, China*

- Built AI Q&A chatbot with RAG retrieval; implemented human handoff system and unknown-query logging
- Built agent to traverse company documents via MCP, filter by format/relevance, and extract to knowledge base
- Built AI field mapping system from Excel to product data sheets with business process detection and automation
- Researched and prototyped LLM observability tools and diagram-preserving RAG for PDFs/Word with figures.

### Math Tutor

Feb. 2025 – Present

*Preply*

*Online*

- Tutored 40+ students in arithmetic through calculus by diagnosing gaps and adapting instruction for mastery
- Earned Super Tutor status and achieved 5-star rating on 700+ lessons through results-driven instruction.

### Lego Engineer Instructor

Sept. 2023 – Feb. 2024

*Level Up Enrichment*

*San Diego, CA*

- Taught 15–20 students per class STEM concepts through hands-on LEGO builds including bridge design
- Designed lesson plans around engineering challenges, fostering problem-solving and teamwork skills

## PROJECTS

### Fog Collector Door Mechanism | CAD, Drivetrain Design, 3D Printing

Professor-Sponsored Research

- Redesigned a 4-motor door system to a single-motor chain-driven mechanism, reducing component costs by 25%
- Prototyped design at 20% scale via 3D printing, then engineered full-scale version for outdoor deployment
- Designed doors to seal collector inlets against environmental debris for reliable outdoor fog water collection

### Sentinel AI | Python, LLM Agents, OpenRouter, OpenClaw

NVIDIA Hackathon

- Built a multi-agentic AI system that simulates, detects, and autonomously investigates cloud security breaches
- Architected 6 specialized LLM agents using NVIDIA Nemotron via OpenRouter and the OpenClaw agent

### Pan-Tilt Tracking Turret | Teensy, OpenCV, Computer Vision

- Built a 2-axis tracking turret with a partner using a Teensy microcontroller, dual motors, and a camera module
- Implemented real-time face, hand, and color detection with OpenCV to calculate target position from live video
- Drove motor control logic in C++ to continuously pan and tilt toward detected targets in real time

### Custom Turbo Boost Gauge | ESP32, GC9B72 Display, Embedded Systems

- Built a real-time turbo boost gauge on ESP32 with circular display and LDR-based day/night mode switching
- Built secondary gauge to toggle IAT and oil temperature modes via button input on a shared SPI bus
- Designed enclosure and mounts in Onshape for thermal management and vibration resistance while driving

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java

**Tools & Frameworks:** ROS 2, OpenCV, Docker, Git, Supabase, PostgreSQL, Agent skills framework

**Hardware:** Teensy, Motors, CAD (Fusion 360 and Onshape), 3D Printing, Raspberry Pi, Arduino Nano, ESP32

**Languages Spoken:** English (Fluent), Cantonese (Conversational), Mandarin (Conversational)

**Activities:** Eagle Scout & Senior Patrol Leader (BSA), Presidential Volunteer Service Award (Gold ×2, Silver, Bronze)

**Interests:** Hiking, Camping, Backpacking, Surfing, Fishing, Martial Arts (12 years), Magic Tricks